

# netneurotools: a trainee-oriented approach to network neuroscience

Received: 24 September 2025

Accepted: 7 August 2026

Published online: 09 September 2026



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Brain imaging is an increasingly interdisciplinary field, encompassing multiple data types and multiple analytic traditions. Projects typically involve many moving parts, such as building customized preprocessing pipelines, transforming between data formats, preparing datasets for analysis and, ultimately, displaying results. The field is conventionally built on highly specialized software packages that solve these individual challenges well but are not necessarily designed to be interoperable. Trainees new to the field are therefore often left to come up with isolated heuristics and workarounds to complete a project. Here we present a set of tools to navigate the increasingly complex informatics ecosystem of brain imaging. netneurotools is a Python toolkit that has been continuously developed and maintained by this laboratory's trainees. The philosophy of the toolkit is that it should be the Swiss army knife of the laboratory: functions and routines that we often use but that are not part of any established pipeline or package. Since its inception, the toolkit has been open and welcomes contributions from neuroscientists across the globe. netneurotools presents a necessary counterweight to out-of-the-box software packages and highlights the importance of smaller, ad hoc functions for implementing projects. By opening a window into the inner workings of a laboratory, netneurotools also presents an opportunity to begin a new type of discourse among groups and establish tangible links within the community.

Modern neuroimaging research requires fluency in many domains, from computing to analytics. Over the past 20 years, numerous software packages have been developed to expedite acquisition, processing, analysis and visualization of imaging data. Software packages that complement current conceptual, technological and analytic trends are more commonly used, whereas others become obsolete. The most enduring packages are often highly specialized, leaving substantial daylight between packages that has to be filled by trainees working on a project.

How does a new trainee, working at the intersection of network neuroscience and brain imaging, get started? A typical project involves

myriad challenges that often encompass multiple established packages as well as custom code: interfacing with dataset application programming interfaces (APIs), implementing algorithms from a wide range of analytical traditions and rendering brain surfaces to display results. Consider the task of community detection (identifying modules in brain connectomes): code to implement the quality function exists in several packages, but the wider framework to conduct the analysis—consensus clustering, mapping the space of degenerate partitions and so on—does not. As a result, common practice is for groups to develop and informally share in-house scripts that rarely survive beyond one or two generations of trainees.

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Here, we present *netneurotools*, a Python utility for implementing those tasks in network neuroscience for which there exists no straightforward solution. *netneurotools* is the internal utility belt of a small laboratory. Initiated at the time of this laboratory's inception at the Montréal Neurological Institute in 2017, the goal of the repository is simply to implement utilities that laboratory members frequently use. By design, functions are diverse and miscellaneous, including functions to construct network null models, estimate graph metrics, fit statistical models, fetch datasets, customize colormaps and plot brain images. The toolbox is 'battle tested': its utilities have been extensively tested and used in more than 15 internal publications. It has also been adopted by the broader neuroscience community. From the beginning, the repository has been open and has seen contributions from all over the world, including ~35 forks and ~120 stars on GitHub.

## Overview and guiding principles

Given the broad use of the repository by our group and others, we have compiled a short report of what the current version of *netneurotools* entails and how it can be used (<https://github.com/netneurolab/netneurotools>). We emphasize that the utilities are intended to fill gaps between large, well-known software packages. Utilities are designed by trainees, for trainees: in this sense, the repository is an organic, living set of utilities that reflects the day-to-day needs of typical trainees working in the field. We close the report with several examples and an outlook for how the repository will continue to develop.

## Practical solutions for filling gaps in network neuroscience workflows

Researchers in network neuroscience are frequently challenged by seemingly 'basic' routines that fall between the boundaries of major software packages. Questions such as how to import a brain atlas, parcellate brain maps, construct a robust consensus network or flexibly visualize results on a brain surface are common yet surprisingly difficult to address with a single tool. Most importantly, researchers need solutions that use simple and straight-forward primitives without requiring

them to learn new frameworks or data formats. Ideally, these solutions should generate reproducible results that can be readily integrated into existing analyses or workflows.

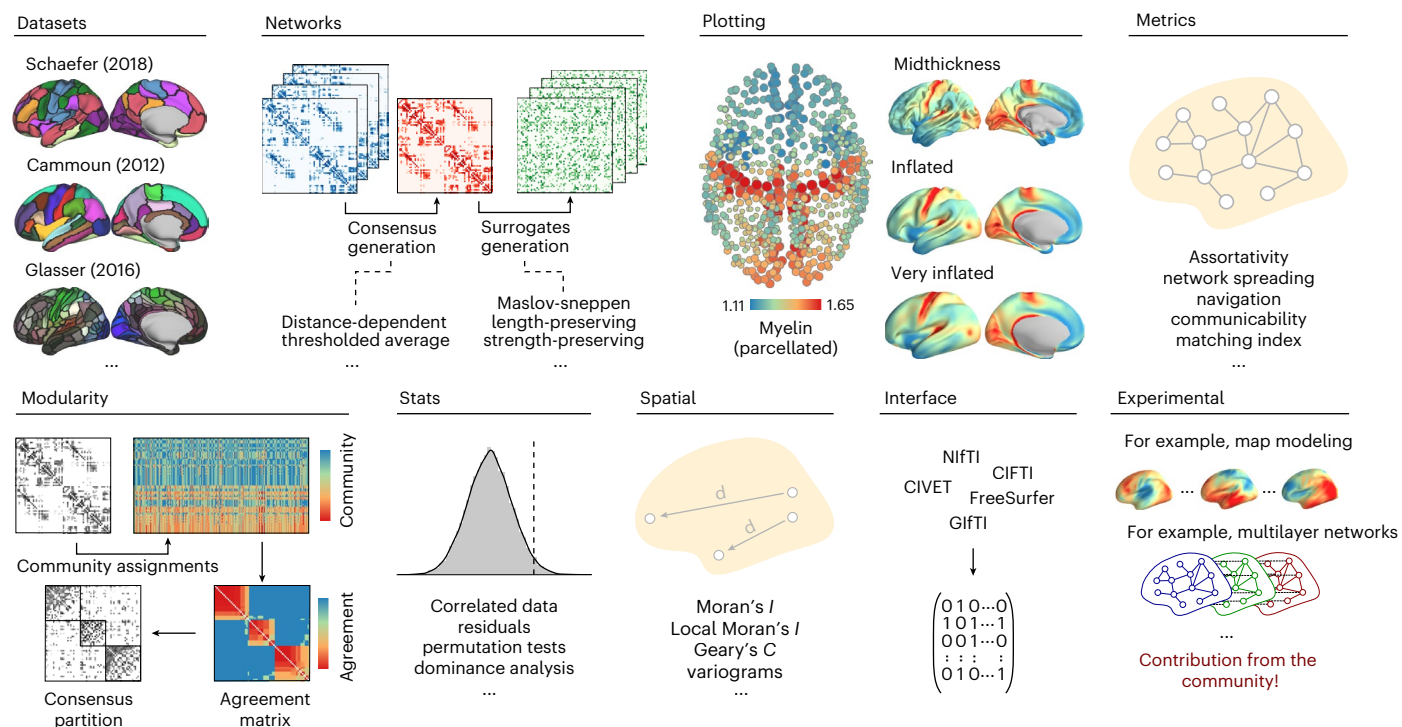
Despite the growing popularity of systems-level analysis, there are few actively maintained and time-tested toolboxes dedicated to addressing these practical challenges in network neuroscience. *netneurotools* fills this gap with a steadily growing collection of functions that aim at lowering barriers and standardizing atomic tasks in network neuroscience research. It builds on 8 years of practical, data-intensive research inquiries by trainees who use these tools daily.

In this sense, the philosophy behind the toolbox is similar to the concept of 'desire paths' in urban design. Here, paths emerge organically as individuals develop preferred routes through urban environments, such as by frequently walking through a park or courtyard. The repeated erosion caused by trampling is taken evidence of a desirable path and subsequently more formally implemented as a walkway by adding gravel or paving. *netneurotools* broadly follows the same idea of structure reflecting function: functionality and workflows are added only after we notice that they are frequently used by laboratory members.

## Toolbox architecture: accessibility and trainee-centric design

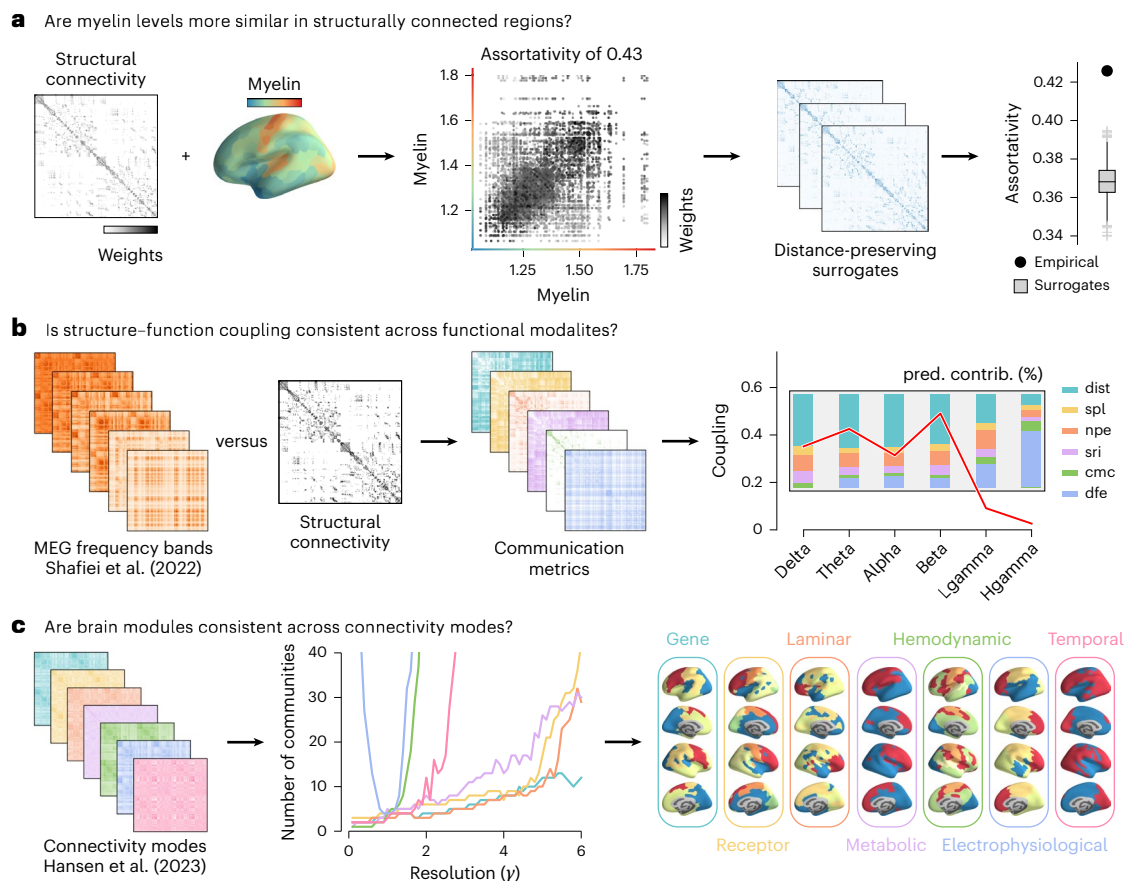
By design, *netneurotools* is trainee-centric and focuses on accessibility and usability. The functions are grouped into intuitive modules, covering the full lifecycle of a project (Fig. 1). A typical network neuroscience project commonly involves data ingestion, network reconstruction, metric estimation, statistical analysis and visualization of results. A researcher starting a project can either bring their own data or simply proceed with the *datasets* module, where there is a large, curated collection of standard templates, common atlases and published datasets for fast and convenient prototyping. The *interface* module can handle potential incompatibilities from external data formats, tools and modalities.

Following data ingestion, the *networks* module allows transforming raw connectivity data into several types of consensus and



**Fig. 1 | A Swiss army knife for network neuroscience.** *netneurotools* implements a versatile and modular set of functions to support a wide range of analyses, from data preprocessing to network metrics and visualization. Functions are

currently grouped in nine distinct modules: datasets, networks, plotting, metrics, modularity, stats, spatial, interface and experimental. See the 'Modules and key functions' section for a brief overview of each module's functionalities.



**Fig. 2 | Answering neurobiologically meaningful questions with**

**netneurotools.** **a.** We provide three tutorial examples for different stages of the research process and explicitly highlight functions that could be used at each stage. In ref. 1, we studied assortative mixing in annotated connectomes. The data used in this study can be directly downloaded from netneurotools using `fetch_bazinet_assortativity`. The assortativity of myelin with respect to the structural connectome can then be quantified using `assortativity_und`. To evaluate the significance of the assortativity coefficient, we can generate surrogate networks that preserve the edge length and degree distribution of the empirical networks with `match_length_degree_distribution`. **b.** In ref. 2, we evaluated structure–function coupling across magnetoencephalography (MEG) frequency bands. We fetched MEG data from<sup>79</sup>, using `fetch_shafiei_megfmrimapping`. Using netneurotools, we can compute communication metrics on the structural connectome (for example, communicability with `communicability_weil`). We can then evaluate the coupling between MEG

connectivity and structural connectivity with a multiple linear regression model using six communication metrics (dist, Euclidean distance; spl, shortest path length; npe, navigation efficiency; sri, search information; cmc, communicability; dfe, diffusion efficiency). To evaluate the contribution of each predictors, we can perform dominance analysis using `get_dominance_stats`. **c.** In ref. 3, we compared the community structure of seven distinct types of interregional similarity matrices (connectivity modes). The data can be downloaded directly from netneurotools using `fetch_hansen_mannynetworks`. To identify the community structure of each connectivity mode for a given resolution  $\gamma$ , we can use `consensus_modularity`, which performs multiple iterations of community detection using the Louvain algorithm<sup>80</sup> and then uses clustering to derive a consensus community assignment, as detailed in ref. 33. Finally, the community assignments can be visualized on the surface of the cortex using `plot_fsaverage_pred_contrib`, predictor contribution.

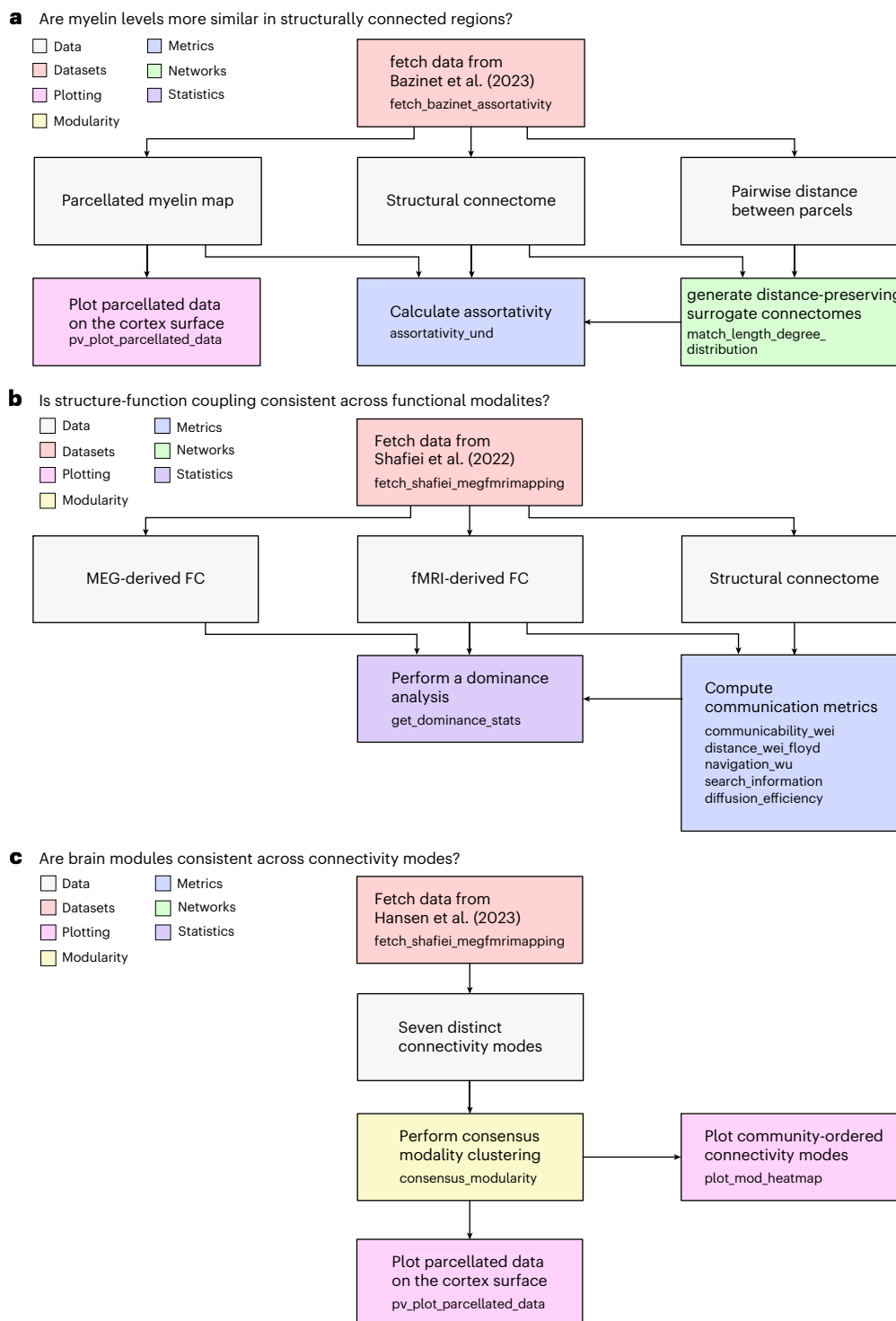
surrogate networks. The networks can then enter `metrics` and `modularity` modules, which provide optimized implementations of classic and cutting-edge network-theoretic metrics, including a streamlined community detection workflow. The `stats` module provides an extensive set of functions for correlations, regressions and permutation tests. The final results can be visualized on the brain surface flexibly with the `plotting` module. By standardizing and streamlining fragmented computational routines into a curated toolbox, netneurotools lets trainees focus more on hypothesis discovery and reproducibility. It also stands as a high-quality, contextualized and trustable reference of routine tasks in network neuroscience, compared with isolated code snippets online and generic responses from large language models.

From datasets to network metrics, from statistics to visualizations, the toolbox mirrors the steps trainees take daily. Adding to the broader usability, functions in netneurotools are modular and transparent and are documented in great detail, allowing flexible mixing and matching.

Trainees can parcellate their own data with curated atlases in `datasets`; pair `metrics` with external graph or machine learning libraries; use `stats` independently for permutation testing and so on.

The toolbox also includes modules adapting to the fast-progressing field. For instance, the `spatial` module that implements geospatial statistics is the most recent addition toward studying spatial embedding and geometric features of the brain. An additional experimental module is available as a sandbox for testing new ideas from the community without disrupting the more stable utilities.

In Fig. 2, we showcase three examples of biologically relevant questions that can be answered using netneurotools functionalities. These questions, inspired from previous work from our group<sup>1–3</sup>, explore whether myelination levels are significantly more similar in structurally connected regions (Fig. 2a), whether structure–function coupling is consistent across functional modalities (Fig. 2b) and whether the brain’s community structure is consistent across modes of connectivity (Fig. 2c). Detailed tutorials can be found in the examples section of



**Fig. 3 | Workflow diagrams. a–c.** Examples of workflow diagrams to answer three questions are provided: are myelin levels more similar in structurally connected regions (a), is structure–function coupling consistent across functional modalities (b) and are brain modules consistent across connectivity modes (c)? Workflow

diagrams illustrating the sequence of analyses presented in Fig. 2. For each analysis, we denote the corresponding netneurotools function. Gray-colored boxes indicate data that can be fetched using the datasets module, whereas the other colors indicate the module in which each function can be found. FC, functional connectome.

the documentation ([https://netneurolab.github.io/netneurotools/auto\\_examples/index.html](https://netneurolab.github.io/netneurotools/auto_examples/index.html)). Workflow diagrams summarizing each analytical step and its associated netneurotools function are presented in Fig. 3.

### Sustainable development and community building

Maintaining scientific software requires concerted effort, especially while supporting an active user base alongside academic

commitments<sup>4</sup>. In the process of keeping netneurotools relevant and publication-ready over time, we have learned several important principles. First, we maintain minimal dependencies and broad applicability. This foundation ensures stability and compatibility with all major platforms. Users can optionally install additional dependencies depending on specific use cases for speed-optimized functions.

Second, we prioritize automated, reproducible builds. For every update and versioned release, netneurotools is automatically built

**Table 1 | Overview of netneurotools modules and representative functions, together with selected examples illustrating their usage within the broader neuroscience community**

| Module       | Description                                                                                         | Key functions                                                                                                                                                                   | Community usage |
|--------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| datasets     | Programmatic download and caching of templates, atlases and neuroimaging datasets                   | <code>fetch_fsaverage</code> , <code>fetch_cammoun2012</code> , <code>fetch_hansen_mannynetworks</code>                                                                         | 54,64,65        |
| networks     | Construction of consensus and surrogate brain networks                                              | <code>func_consensus</code> , <code>struct_consensus</code> , <code>randmio_und</code> , <code>threshold_network</code>                                                         | 66–69           |
| metrics      | Graph-theoretic measures including communication dynamics and statistical network metrics           | <code>distance_wei_floyd</code> , <code>navigation_wu</code> , <code>communicability_wei</code> , <code>assortativity_und</code> , <code>network_pearsonr</code>                | 70              |
| modularity   | Community detection and assessment of consensus partitions                                          | <code>consensus_modularity</code> , <code>find_consensus</code> , <code>get_modularity</code>                                                                                   | 71,72           |
| stats        | Statistical inference including correlation, regression, permutation testing and dominance analysis | <code>efficient_pearsonr</code> , <code>permtest_1samp</code> , <code>permtest_pearsonr</code> , <code>get_dominance_stats</code>                                               | 73–76           |
| spatial      | Spatial statistics metrics for studying the spatial organization of neural patterns                 | <code>morans_i</code> , <code>local_morans_i</code> , <code>gearys_c</code> , <code>lees_l</code>                                                                               |                 |
| plotting     | Publication-quality visualization of brain data                                                     | <code>pv_plot_surface</code> , <code>pv_plot_parcellated_data</code> , <code>plot_point_brain</code> , <code>plot_mod_heatmap</code>                                            | 77,78           |
| interface    | Parsers and converters for external neuroimaging tools and file formats                             | <code>extract_cifti_surface</code> , <code>extract_annot_labels</code> , <code>extract_gifti_labels</code> , <code>vertices_to_parcel</code> , <code>parcels_to_vertices</code> |                 |
| experimental | Sandbox for incubating community-contributed methods before stable integration                      | Community-contributed functions, for example, multilayer network analysis and surrogate brain map generation                                                                    |                 |

with GitHub Actions as Docker containers. Updates are automatically tested to ensure that core functionality remains reliable. A continuously updated documentation is similarly compiled and hosted on GitHub Pages (<https://netneurolab.github.io/netneurotools/>), providing detailed function references, examples and tutorials.

Third, we encourage sustainable community involvement both internally and externally. Within our laboratory, trainees are encouraged to reuse and improve the existing functions, as well as incorporating new functions from their projects in netneurotools. New functions are contributed through pull requests, reviewed by maintainers and integrated only after passing automated continuous integration tests. Depending on the function type, validation may include benchmarking against analytical expectations, reproducing outputs from established implementations in other software ecosystems (for example, MATLAB or R), or reproducing published findings from peer-reviewed studies.

Externally, we actively collaborate with researchers worldwide to incorporate new functions and address community needs through a support ticket system. The research-driven development ensures that the toolbox stays updated with the frontier questions and methods in network neuroscience and brain imaging. We also actively foster an open science environment to enable continual transfer of knowledge transcending the typical graduate student lifecycle. Functions in netneurotools originally developed by early laboratory members are now being maintained and extended by newer generations of trainees.

## Modules and key functions

### The netneurotools package

The netneurotools package is organized into nine modules that provide comprehensive functionality for data acquisition, network construction, graph theory calculations, visualization, statistical analysis and interfacing with external neuroimaging tools. See Table 1 for an overview of each module, with representative functions and selected examples of publications illustrating their usage within the broader neuroscience community. netneurotools is distributed under the BSD-3 license, making it free for academic and commercial use. The package is available on GitHub (<https://github.com/netneurolab/netneurotools>) with comprehensive documentation (<https://netneurolab.github.io/netneurotools/>) and tutorials ([https://netneurolab.github.io/netneurotools/auto\\_examples/index.html](https://netneurolab.github.io/netneurotools/auto_examples/index.html)). It can be installed via

PyPI (`pip install netneurotools`), from source or used from a Docker container.

### Toolbox dependencies

netneurotools depends on a minimal set of scientific Python packages: NumPy<sup>5</sup>, SciPy<sup>6</sup>, Scikit-learn<sup>7</sup> and NetworkX<sup>8</sup> for numerical procedures and NiBabel<sup>9</sup> and Nilearn<sup>10</sup> for routine neuroimaging operations. The package leverages Joblib<sup>11</sup> for parallelization and Numba<sup>12</sup> for just-in-time compilation of computationally intensive functions. For visualization, netneurotools uses Matplotlib<sup>13</sup> for two-dimensional figures and PyVista<sup>14</sup>, a streamlined interface to the Visualization Toolkit (VTK)<sup>15</sup>, for publication-quality three-dimensional renderings of brain surfaces.

### Ingesting data: the datasets and networks modules

The netneurotools.datasets module provides an integrated infrastructure for programmatically downloading, loading and managing various neuroimaging datasets, atlases, templates and published results. Key functions include fetching cortical surface templates (for example, `fetch_fsaverage`<sup>16</sup>, `fetch_cont69`<sup>17</sup>), brain parcellations (for example, `fetch_cammoun2012`<sup>18</sup>, `fetch_schaefer2018`<sup>19</sup>) and published datasets (for example, `fetch_vazquez_rodriguez2019`<sup>20</sup>). The module implements a caching system to efficiently store and retrieve downloaded data. The netneurotools.networks module contains methods for generating and preprocessing brain networks. It includes functions for creating consensus functional (`func_consensus`) and structural (`struct_consensus`<sup>21</sup>) connectivity matrices from a group of subjects, generating randomized networks while preserving desired topological or geometrical properties (`randmio_und`<sup>22</sup>, `strength_preserving_rand_sa`<sup>23</sup>) and thresholding networks (`threshold_network`, `binarize_network`).

### Visualizing the brain: the plotting module

The netneurotools.plotting module offers routines for creating publication-ready visualizations of volumetric, surface and network-structured brain data. Surface-based plotting functions (`plot_fsaverage` and `pv_plot_surface`) allow convenient and flexible mapping of cortical surfaces, with customizable views, lights and colors. Coordinate-based plotting function (`plot_point_brain`) renders

brain regions as three-dimensional point clouds. This module also provides specialized functions for visualizing network communities (`plot_mod_heatmap`).

### Calculating graph metrics: the `metrics` and `modularity` modules

The `netneurotools.metrics` module features an optimized collection of functions for working with brain networks, implementing both classical and novel graph theoretical measures<sup>24</sup>. These include functions for calculating basic nodal properties (`degrees_und` and `degrees_dir`), shortest path and routing metrics (`distance_wei_floyd`<sup>25–27</sup> and `navigation_wu`<sup>28</sup>), communication dynamics (`communicability_bin`<sup>29</sup>, `communicability_wei`<sup>30</sup> and `mean_first_passage_time`) and statistical network metrics (`network_pearsonr`<sup>31</sup> and `network_variance`<sup>32</sup>). Many implementations offer substantial performance improvements over existing packages through vectorization and Numba acceleration.

The `netneurotools.modularity` module includes functions for exploring and analyzing community structure in brain networks. It provides tools for identifying consensus community assignments across multiple partitioning solutions (`find_consensus`<sup>33</sup>), assessing the quality of community assignments (`get_modularity`, `get_modularity_z`) and calculating partition similarity (`zrand`<sup>34</sup>).

### Estimating spatially naive and spatially embedded statistics: the `stats` and `spatial` modules

The `netneurotools.stats` module provides functions for statistical analysis and inference in neuroimaging. It includes efficient implementations of correlation metrics (`efficient_pearsonr` and `weighted_pearsonr`), permutation testing routines (`permtest_1samp` and `permtest_pearsonr`) and regression analyses (`residualize` and `get_dominance_stats`<sup>35,36</sup>).

The `netneurotools.spatial` module features functions for studying the spatial organization of neural patterns, offering tools to calculate spatial autocorrelation metrics (Moran's  $I$ <sup>37–39</sup>, Geary's  $C$ <sup>40,41</sup> and Lee's  $L$ <sup>42</sup>) and their local variants. These functions enable researchers to study the spatial dependence of structural and functional brain features.

### Navigating complex tools: the `interface` module

The `netneurotools.interface` module includes functions for interacting with external neuroimaging tools and file formats. It provides parsers and converters for FreeSurfer<sup>43</sup> (`extract_annot_labels`), CIFTI<sup>44</sup> (`extract_cifti_surface`) and GIFTI formats, facilitating the extraction and manipulation of data from these complex file formats, enabling seamless workflow integration with established neuroimaging software packages.

### Incubating new ideas: the `experimental` module

The `netneurotools.experimental` module allows users to experiment with cutting-edge methods before they are fully integrated into the main package. It is a sandbox and incubator to try out functions that are novel but still unstable. It thus facilitates and encourages new contributions from the community, which could include, for example, new methods for multilayer network analysis or for surrogate brain map generation.

### Automated testing and release

`netneurotools` uses continuous integration through GitHub Actions to ensure code quality and reliability. The automated process runs comprehensive unit tests across multiple Python versions and operating systems upon code changes, ensuring that new contributions maintain compatibility and functionality. The package is regularly released and published to PyPI and Docker Hub, making it readily available to the scientific community.

### Relationship with existing packages

`netneurotools` occupies a complementary niche relative to established software in network neuroscience. By design, it operates on regionalized data represented in network format: parcellated brain maps and connectivity matrices. It focuses on analytical steps that follow initial network reconstruction and does not handle the preprocessing of raw imaging data typically managed by upstream tools for functional preprocessing (for example, fMRIPrep<sup>45</sup> and CONN<sup>46</sup>) and diffusion tractography (for example, QSIprep<sup>47</sup>, MRtrix<sup>48</sup> and DSIStudio<sup>49</sup>) pipelines. Likewise, large professionally maintained packages such as Nilearn<sup>10</sup>, NiBabel<sup>9</sup> and Connectome Workbench (<https://www.humanconnectome.org/software/connectome-workbench>) provide robust infrastructure for neuroimaging data management and surface processing at the volume and voxel level and serve as important pillars of the ecosystem; users are directed to these toolboxes for native operations at those levels. As a result, networks analyzed with `netneurotools` will inherit the analytical variability that arises from upstream preprocessing and connectivity matrix construction<sup>50,51</sup>; it is therefore the responsibility of the user to standardize these steps and assess their impact on downstream analyses.

Within the scope of network-level analysis, several toolboxes offering end-to-end graph-theoretic pipelines (for example, GREYNET<sup>52</sup>) or Python ports of established MATLAB libraries (for example, `bctpy` (<https://github.com/aestrivex/bctpy>) and `brainconn` (<https://github.com/fiuneuro/brainconn>)) have seen reduced maintenance activity and do not fully leverage modern scientific Python performance improvements. `netneurotools` draws on Brain Connectivity Toolbox<sup>24</sup> and NetworkX<sup>8</sup> as references for graph-theoretic and general network analysis, providing optimized Python-native implementations of select measures alongside original methods not available in either. The top-down development model and broad scope of larger packages are intrinsically different from the trainee-driven, organic iteration that characterizes day-to-day research.

On the visualization side, two-dimensional region-specific packages (for example, Simple-Brain-Plot<sup>53</sup> and `subcortex_visualization`<sup>54</sup>) are quick and convenient for limited use cases, whereas legacy options in MATLAB or unmaintained Python libraries (for example, BrainNet Viewer<sup>55</sup> and `PySurfer` (<https://github.com/nipy/PySurfer/>)) leave a gap for flexible, publication-quality surface rendering across cortical and subcortical structures.

Other tools are highly specialized in a single analytical direction: `neuromaps`<sup>56</sup> in multimodal brain map contextualization, NBS<sup>57</sup> in network-based statistics, Comet Toolbox<sup>58</sup> in dynamic functional connectivity and ENIGMA Toolbox<sup>59</sup> and Brainspace<sup>60</sup> in normative and gradient analyses. Relative to these toolboxes, `netneurotools` is notable for integrating cross-domain breadth, spanning data ingestion, network construction, statistical inference, spatial analysis and visualization within a single interoperable Python package and for including original methods derived directly from peer-reviewed publications. The small, modular, decoupled functions are agnostic to upstream preprocessing choices, making integration into existing pipelines straightforward. More broadly, the toolkit embodies a model of sustainable research software development: rather than being designed around a prespecified scope, `netneurotools` has grown organically from the day-to-day needs of the laboratory, remaining active precisely because it is built and maintained by the same trainees who rely on it, in contrast to more ambitious toolboxes that have lost traction after initial release.

### Conclusion

The challenge of accessible and reproducible research practices is substantial and ongoing. `netneurotools` not only brings tools of the trade but also embodies our efforts of implementing findable, accessible, interoperable and reusable (FAIR) principles<sup>61</sup> in research coding<sup>62,63</sup>. By curating standardized procedures, facilitating data sharing and ensuring code transparency, we hope `netneurotools` lowers the barrier for entry into, and raises rigor in, the field of network neuroscience.

As the field continues to grow, so too will netneurotools. We envision it as an evolving set of utilities that reflect the day-to-day needs of both new and experienced trainees. Through research-driven, sustainable development, we aim to ensure that netneurotools remains a useful resource for the community: bridging gaps between specialized tools and empowering researchers to focus on scientific discovery rather than technical implementation.

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## Acknowledgements

We thank Y. Zhou, M. Pourmajidian and T. Taheri for comments and suggestions on the manuscript. We also thank past and future contributors to the toolbox. B.M. acknowledges support from the Natural Sciences and Engineering Research Council of Canada (NSERC), Canadian Institutes of Health Research (CIHR), Brain Canada Foundation

Future Leaders Fund, the Canada Research Chairs Program, the Michael J. Fox Foundation, and the Healthy Brains for Healthy Lives initiative. Z.-Q.L. acknowledges support from the Fonds de Recherche du Québec – Nature et Technologies (FRQNT). V.B. acknowledges support from the Natural Sciences and Engineering Research Council of Canada (NSERC), the Fonds de Recherche du Québec – Nature et Technologies (FRQNT), the Healthy Brains for Healthy Lives initiative and Brain Canada.

### Author contributions

Z.-Q.L., V.B. and B.M. conceived the study and wrote the manuscript, with valuable revision from all of the authors. All the authors contributed to developing the software toolbox. B.M. was the project administrator.

### Competing interests

The authors declare no competing interests.

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**Peer review information** *Nature Protocols* thanks Yanchao Bi, Hamza Farooq, Dingchen Zhang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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